Sahaj Bhadja

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Summary

Software engineer with 2+ years building Python APIs, automation workflows, and data applications across FastAPI, Flask, AWS, SQL, CI/CD, debugging, and production documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built ETL-style integration pipelines with FastAPI, Supabase, SQL Server, and cloud storage, processing 5K+ monthly jobs with reconciliation checks, error handling, and CI/CD-backed releases, achieving 99%+ production reliability.
- Restructured a fault-tolerant async orchestration service using FastAPI background workers, processing 1K+ AI jobs with explicit job-state transitions and failure isolation, improving throughput 3x under concurrent workloads.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented a production Flask backend on AWS serving 150+ instructors and 10K+ daily grading requests, with CI/CD-driven rubric workflows, secure file ingestion, and stable API performance during peak load.
- Designed a concurrent batch-processing pipeline using multithreading, OCR, and REST APIs to process 1K+ uploaded files per batch, improving debugging clarity and maintaining fault-tolerant execution across grading workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform data-sharing software with automated VM testing, executable builds, and 20+ unit and integration suites using Jenkins, PyTest, and Mockito, strengthening release reliability across three operating systems.
- Troubleshot and optimized large-scale API integrations with AWS Glue and S3, improving response times by 30% while leading ETL workflow improvements that boosted pipeline efficiency 68% and reduced technical debt 20%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based resource management platform with WebSockets, REST APIs, and microservices, supporting six core workflows and 2M+ synchronized records while reducing code-review turnaround 25% across release cycles.
- Automated deployments with Jenkins CI/CD and Docker-based packaging, achieving 95%+ successful build validation across release runs and cutting deployment time by 50% for staging-ready application updates teamwide.

LST Capitals **Los Angeles, CA**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs for booking management, pricing, coupon validation, payments, and real-time status updates across 50K+ booking records, maintaining sub-200ms responses across mobile and web reservation workflows.
- Integrated a PCI-compliant multi-currency payment infrastructure with webhook automation, refund handling, and analytics dashboards, reducing manual payment operations by 70% while improving observability for cross-functional business teams.

Skills

Languages: Python, SQL, JavaScript, TypeScript, Java, C++, Bash, Shell
Backend & Cloud: FastAPI, Flask, Django, REST APIs, Node.js, Express.js, WebSockets, AWS EC2, AWS S3, AWS RDS, AWS Glue, ECS/Fargate, Azure SQL, Docker, Linux, NGINX
Automation & Quality: Ansible, Jenkins, GitHub Actions, PostgreSQL, MySQL, MongoDB, Firebase, PyTest, CI/CD, ETL, automation testing, performance optimization, debugging, documentation, code reviews, Postman, Git, Agile/Scrum

Projects

Linguistix Code | *Python, AWS, Docker*
- Developed and deployed a containerized RESTful data processing API on AWS via CodePipeline and ECS/Fargate, with automated CI/CD releases, validation checks, and 87% accuracy for recurring ingestion workflows.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Engineered a cloud ingestion and indexing pipeline processing 50K+ records, implementing incremental updates, metadata persistence, and performance tuning to reduce retrieval latency by 40% for semantic search workloads.

HMIS Live | *MERN, ABDM APIs, Twilio, Sentry*
- Implemented consent-based patient-history access, appointment alerts, monitoring, and structured API workflows for a healthcare platform, reducing administrative workload by 30% while supporting secure clinician access to real-time records.